higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING
NQF LEVEL 2

1 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1: INTRODUCTION TO CIVIL AND STRUCTURAL STEEL WORK

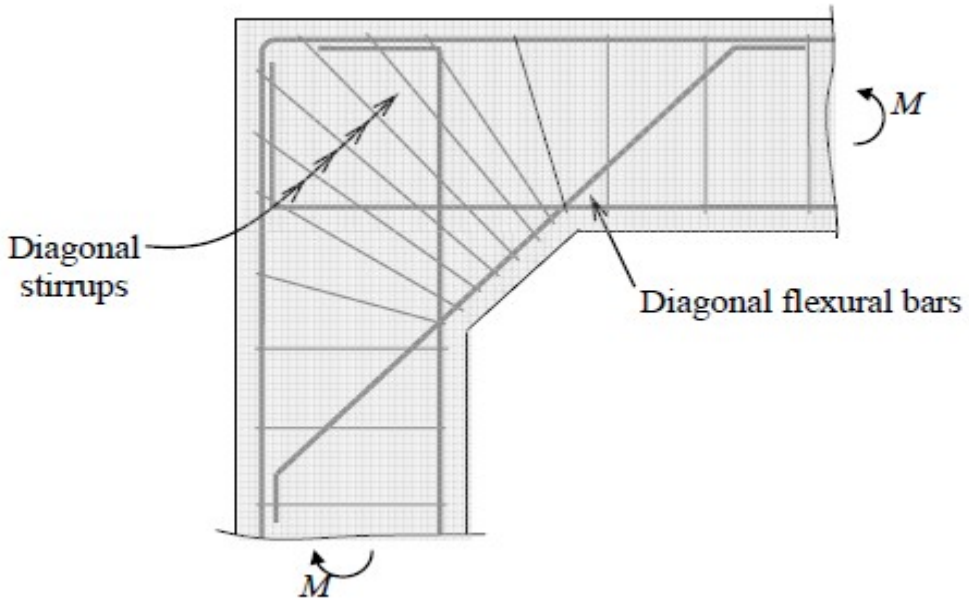
1.1 1.1.1 Portal frames are two-dimensional rigid frames that have the basic characteristics of a rigid joint between a column and a beam. (2)

1.1.2 The main objective of this form of design is to reduce bending moment in the beam, which allows the frame to act as one structural unit. (2)

1.2 Corrosion is the formation of compounds when exposed to air (oxygen), water or electrolytes such as salt water. (3)

1.3 A legend is sometimes referred to as the alphabet of lines. Line style can provide a lot of information to the user. (2)

1.4



Component	Mark allocation
Quality of line work	2
Diagonal stirrups	4
Diagonal flexural bars	1
Other rebars	4
Title block	1
Scale	1
General impression	2

(15)

1.5 High-tensile steel has relatively high carbon content and a high tensile strength, while mild steel is made up of iron with less than 0,3% carbon and it can be easily cut, bent and welded. (2 + 2) (4)

1.6 It is a wall built to hold back (retain) soil. (2)

1.7 The client must first access funds and the land on which to build. (2)

1.8 Main contractors enter into a contractual agreement with the client/owner to build a building according to the design specifications, while subcontractors are employed by the main contractor to do specific parts of the building project. (3)

1.9
- Drawing number and revision suffix
- Project number
- Scale, date of drawing, names or initials of the persons who designed, drew and checked the drawing
- Project title
- Drawing title
- Name, address and telephone number of the organisation (Any 5 × 1) (5)

[40]

QUESTION 2: STRUCTURAL STEEL WORK AND REINFORCED CONCRETE

2.1
- Cement is the binding material.
- Coarse aggregate consists of crushed stones and gives mass to the concrete and prevent shrinkage of cement.
- Fine aggregate consists of river sand and prevents shrinkage of cement.
- Clean water is used to make concrete, it activates the hydration of cement and forms plastic mass. (Any 2 × 2) (4)

2.2
- Fire resistance
- Durability
- Strength (3)

2.3

Member ✓	Bar mark✓	Type and size✓	No. of members✓	Length of each bar✓	Shape✓
BEAM✓	01✓	Y20✓	3✓	2 200✓	
BEAM	02✓	Y16✓	2	2 200	
BEAM	03	R10✓	10✓		

(15)

2.4
2.4.1 durability
2.4.2 deterioration
2.4.3 weathering
2.4.4 chemical
2.4.5 wind
2.4.6 rain

(6 × 1) (6)

2.5
- Pour a 50 mm layer of concrete into a mould and temper the concrete by using temping rods with exactly 45 blows.
- Repeat this step three times until the mould is completely filled.
- Use a hand trowel to level the concrete at the top of the mould.
- Label the mould with the date, the mixer's name and number of the mould.
- Cover the mould with a damp cloth.
- Fill at least six moulds with concrete. (6)

2.6 Stress is the amount of force that is exerted on steel, while strain refers to how much steel deforms when that force is applied to it. (2)

2.7 2.7.1 Ability to carry a load
2.7.2 Deforming without failing
(2 × 2) (4)
[40]

QUESTION 3: BOLTS IN STRUCTURAL STEEL WORK

3.1
- Unskilled labour
- Noiseless and quick fabrication
- Fast progress of work
- No special equipment needed for installation (4)

3.2 High-strength friction-grip (HSFG) bolts clamp joint surfaces together so that they can transfer loads by friction between the surfaces. (2)

3.3 Holding-down bolts are fabricated from a round bar that complies with the requirements of SANS 1431 grade 300WA. (2)

3.4
- Not exceeding 24 mm diameter: 2 mm greater than the bolt diameter
- Greater than 24 mm diameter: 3 mm greater than the bolt diameter (2)
[10]

QUESTION 4: WELDING SYMBOLS AND JOINTS

4.1 C (1)

4.2 It is a type of joint formed when the two pieces to be joined lie in parallel planes and their edges overlap. (2)

4.3 4.3.1 Primary element of any welding symbol
4.3.2 Graphic description of the type of weld
4.3.3 First dimension appearing to the immediate right of the weld symbol
(3 × 1) (3)

4.4 4.4.1 T, corner
4.4.2 Unlimited thickness, complete joint penn.
4.4.3 Backing bar
4.4.4 Single-bevel groove

(4 × 1) (4)
[10]

TOTAL: 100